



EUNOIA JUNIOR COLLEGE
JC1 Promotional Examination 2020
General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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REGISTRATION
NUMBER

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CHEMISTRY

Paper 2 Structured Questions and Free Response

9729/02

01 October 2020

2 hours

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Do not use paper clips, highlighters, glue or correction fluid.

Write your name, civics group, registration number on all the work you hand in.

Answer all questions in the spaces provided on this Question Paper.

Section A

You are advised to spend 1 hour 30 minutes on this section.

There are five questions in this section.

Answer **all** questions in this section.

Section B

You are advised to spend 30 minutes on this section.

There are two questions in this section. **Choose 1 out of the 2 questions.**

Write all answers in **dark blue or black pen.**

You may use an HB pencil for any diagrams or graphs.

The number of marks is given in brackets [] at the end of each question or part question.

A Data Booklet is provided.

The use of an approved scientific calculator is expected, where appropriate.

For Examiner's Use

Paper 2

Section A

1 /10

2 /17

3 / 8

4 /12

5 /13

Section B

6 or 7 /20

P2 Total /80

Paper 1 /30

Paper 4 /33

Section A

Answer **all** the questions in this section.

1 *Use of the Data Booklet is relevant to this question.*

Chromium has two naturally occurring isotopes, ^{52}Cr and ^{54}Cr . Some common oxidation states of chromium are +2 and +3.

(a) Write the full electronic configuration of Cr.

Cr: [1]

(b) Explain why the 2nd ionisation energy of Cr is lower than its 3rd ionisation energy.

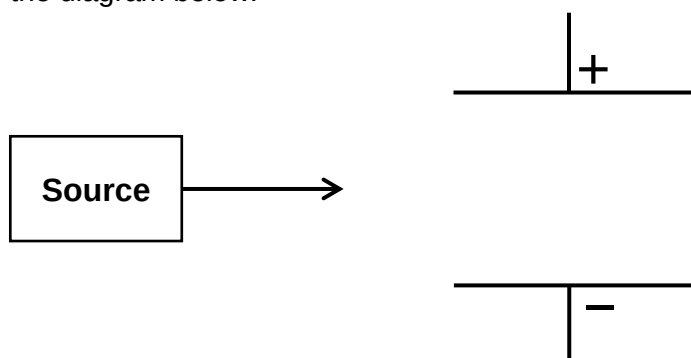
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..... [2]

(c) When passed through an electric field of a certain charge, $^{52}\text{Cr}^{2+}$ ion deflects by an angle of 6° .

(i) Calculate the angle of deflection of $^{54}\text{Cr}^{3+}$ ion when passed through the same electric field.

[1]

- (ii) Sketch clearly how $^{54}\text{Cr}^{3+}$ ion will behave when passing through an electric field in the diagram below.



[1]

- (d) Chromium can form oxides such as chromium(IV) oxide, CrO_2 .

In the presence of dilute sulfuric acid, CrO_2 will disproportionate to Cr^{3+} and $\text{Cr}_2\text{O}_7^{2-}$.

- (i) By constructing the half equations, write the balanced equation for the disproportionation of CrO_2 in acidic solution.

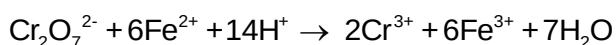
Oxidation:

Reduction:

Overall:

[2]

- (ii) The resulting solution from (d)(i) needed 10.00 cm^3 of $0.0200\text{ mol dm}^{-3}\text{ Fe}^{2+}$ to completely reduce $\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+} .



Use the data to calculate the mass of CrO_2 that has undergone disproportionation.

[3]

[Total: 10]

- 2 Hydrazine, N_2H_4 , is an inorganic compound largely used as a foaming agent in the preparation of polymer foam. It is manufactured via the Olin Raschig Process where aqueous sodium hypochlorite, NaClO , is mixed with ammonia, NH_3 , to produce anhydrous hydrazine.

(a) Draw the dot-and-cross diagram of NaClO .

[2]

A kinetic study has been conducted to determine the order of reaction of the reacting species and the kinetics mechanism.

(b) In one experiment, the following results were obtained.

time / min	$[\text{ClO}^-] / \text{mol dm}^{-3}$, where $[\text{NH}_3] = 0.10 \text{ mol dm}^{-3}$
0	0.0100
30	0.0079
60	0.0062
90	0.0049
120	0.0038
150	0.0030

- (i) Using the data above, plot the graph of $[\text{ClO}^-]$ against time on the grid in Fig. 2.1.

[2]

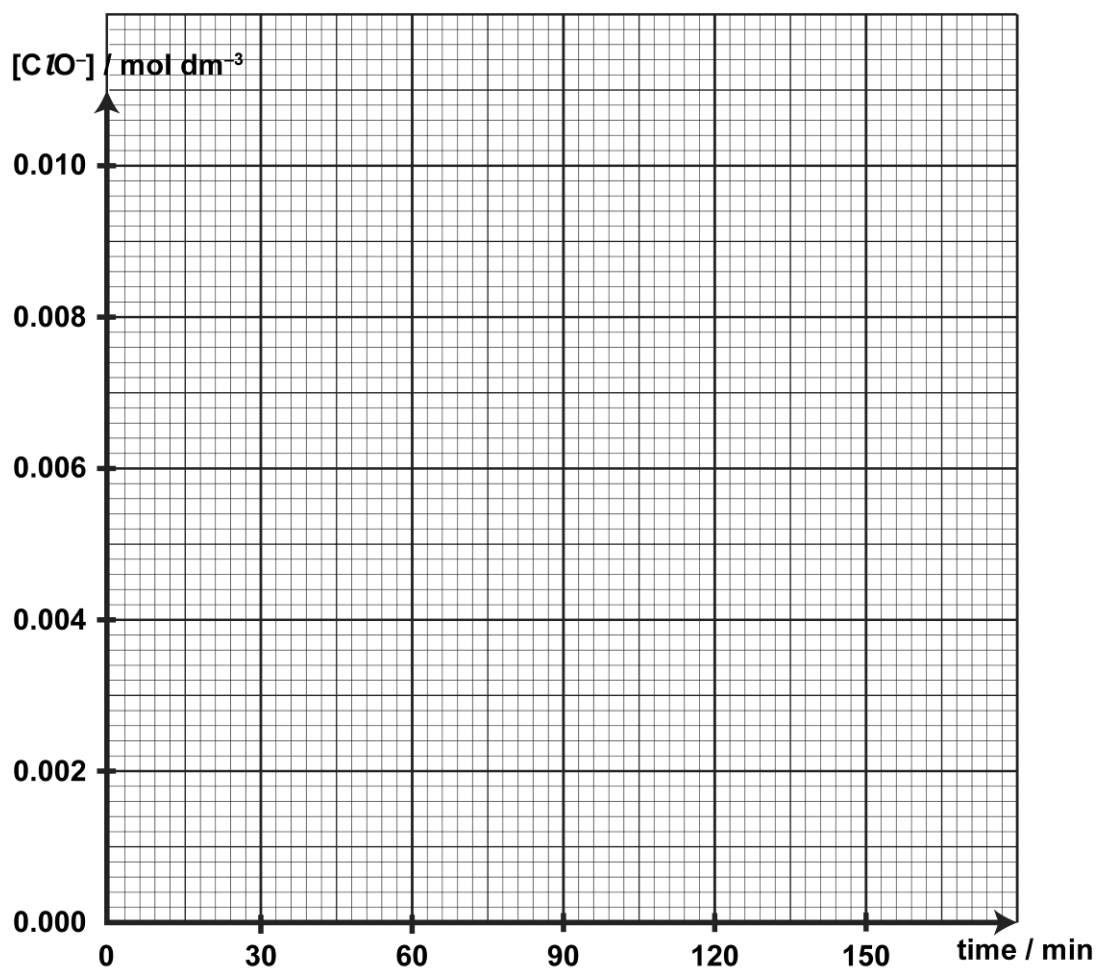


Fig. 2.1

(ii) Use your graph to determine the order of reaction with respect to ClO^- .

.....
.....
..... [2]

- (c) In another series of experiments, the time taken for a fixed volume of hydrazine to be produced was monitored and the following data was obtained.

Table 2.1

experiment	$[\text{ClO}^-] / \text{mol dm}^{-3}$	$[\text{NH}_3] / \text{mol dm}^{-3}$	time taken / min
1	0.10	0.10	12.5
2	0.10	0.15	8.33
3	0.15	0.05	

- (i) Deduce the order of reaction with respect to NH_3 . Write the rate equation and the units of the rate constant.

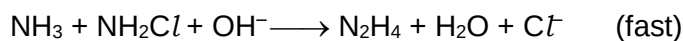
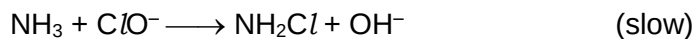
[4]

- (ii) Hence, fill in the time taken for experiment 3 to be completed in Table 2.1.

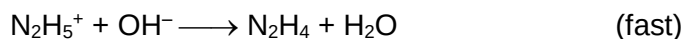
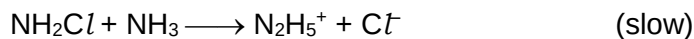
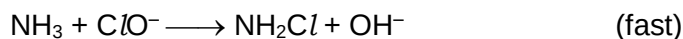
[1]

(d) From kinetic studies, the following mechanisms were proposed.

Mechanism A:



Mechanism B:



(i) Explain which of the two mechanisms is consistent with the kinetic data.

.....

 [2]

(ii) Hence, write the overall equation for the reaction between sodium hypochlorite and ammonia to produce hydrazine.

..... [1]

(e) The Olin Raschig Process was modified from the Raschig Process used to produce liquid hydroxylamine, NH_2OH , in the presence of copper metal.

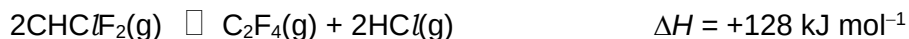
Explain how copper acts as a heterogeneous catalyst in this process.

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 [3]

[Total: 17]

- 3 Upon strong heating in the absence of air, pyrolysis of chlorodifluoromethane, CHClF_2 , occurs according to the following equilibrium.



At 450°C , a closed vessel with a volume of 18.5 dm^3 , contains 1.00 mol of CHClF_2 . When equilibrium was reached, the amount of CHClF_2 in the vessel was found to be 0.200 mol .

- (a) Calculate the K_c for the reaction at 450°C .

[2]

- (b) Explain how the equilibrium amount of C_2F_4 would change if the experiment is repeated in a larger vessel at 450°C .

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..... [2]

(c) When the reaction was repeated at a different temperature, the K_c of the reaction was found to have a magnitude of 0.780.

(i) Given that the degree of dissociation of CHCl_3 is 75%, calculate the equilibrium concentration of HCl at the new temperature.

[3]

(ii) By considering your answer in (a), state whether the pyrolysis process in (c)(i) was repeated at a higher or lower temperature.

..... [1]

[Total: 8]

- 4 Butane, C_4H_{10} , is a gas that is commonly found in canisters used in portable gas stove.

Some properties of butane are given in Table 4.1 as shown.

Table 4.1

relative molecular mass	58.1
density at 15 °C	2.48 kg m ⁻³
standard enthalpy change of combustion	-2878 kJ mol ⁻¹

- (a) Define the term *standard enthalpy change of combustion* of butane.

.....

 [1]

- (b) During winter, a family used a canister containing 225 g of butane to heat up 3 litres of ready-made soup. It took 10 min for the soup to be heated from its initial temperature of 15 °C to reach its boiling point of 100 °C.

- (i) Calculate the volume, in cm³, occupied by 225 g of butane at 15 °C.

[1]

- (ii) The volume of the butane canister used is 500 cm³. Explain the difference between this value and your answer in (b)(i).

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 [1]

- (iii) The combustion process was known to be only 60% efficient. Calculate the mass of butane used to heat the soup from its initial temperature to its boiling point. (Assume specific heat capacity of soup = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$)

[3]

- (iv) Suggest a reason why the combustion of butane is not 100% efficient.

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..... [1]

- (c) Under ultraviolet light, butane reacts with bromine to form 2-bromobutane. Describe the mechanism for this reaction.

[3]

- (d) Butane can be formed by the hydrogenation of but-1-ene, which has a molecular formula of C_4H_8 .

- (i) State the type of reaction but-1-ene has undergone.

..... [1]

- (ii) Under ultraviolet light, another isomer of C_4H_8 , reacts with bromine to form 3 possible mono-brominated products (ignore stereoisomers).
Draw the **skeletal** formula of this isomer.

[1]

[Total: 12]

5 Halogen compounds have a wide range of uses and are commonly used as precursors for conversion into other compounds.

(a) The boiling point of Br_2 is 59°C , while the boiling point of I_2 is 184°C .

(i) Explain the difference in boiling point between the two halogens.

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..... [1]

(ii) Bromine and iodine react to form the interhalogen compound IBr .

Discuss the relative strength of the different types of intermolecular forces of attraction between IBr molecules, given the boiling point of IBr is 116°C .

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..... [2]

(b) When but-1-ene is treated with IBr , the major product, 2-bromo-1-iodobutane, is formed.

(i) Draw a labelled diagram to show how the orbitals overlap to form the $\text{C}=\text{C}$ bond in an alkene, and state the type of hybridisation involved.

[2]

(ii) Name the mechanism of the reaction between IBr and but-1-ene.

..... [1]

(iii) Explain why the major product is 2-bromo-1-iodobutane and not 1-bromo-2-iodobutane, in terms of

- which atom in IBr first reacts with but-1-ene to form the intermediate
- structure of all possible intermediates formed during the reaction.

You may wish to include in your answer the structure of the possible intermediates.

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..... [2]

(iv) State the type of stereoisomerism displayed by 2-bromo-1-iodobutane and draw the structure of the isomers.

Type of stereoisomerism

[2]

(c) **A** and **B** are structural isomers with the molecular formula C_4H_7Br .

A is formed from 2-bromo-1-iodobutane and decolourises aqueous bromine.

B does not decolourise aqueous bromine and exhibits stereoisomerism.

Deduce the structures of **A** and **B**. State the reagents and conditions required to convert 2-bromo-1-iodobutane to **A**.

A	B
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Reagents and conditions [3]

[Total: 13]

Section B

Answer **one** question from this section.

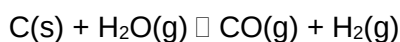
- 6 (a) State what is meant by the term *dynamic equilibrium*. [2]

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- (b) Water gas, a mixture of H_2 and CO , can be formed by reacting steam with excess powdered carbon.



- (i) At 1000 K, the equilibrium partial pressure of CO(g) is determined to be 5.40 atm in a sealed vessel of 1 dm^3 . Assuming ideal gas behaviour, calculate the minimum mass of carbon required to establish equilibrium under such conditions. [2]
- (ii) In another experiment, 1 mol of each gas, $\text{H}_2\text{O(g)}$, CO(g) and $\text{H}_2\text{(g)}$ was subjected to high pressure. CO(g) was observed to deviate most from ideal gas behavior. Suggest a reason for the observation. [1]

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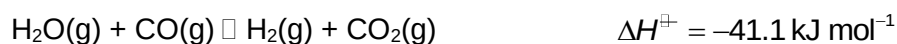
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(c) The CO(g) in water gas can be further reacted in the water gas shift reaction (WGSR).



A sample of water gas containing an equimolar mixture of CO(g) and H₂(g), as well as 0.600 atm of H₂O(g) was introduced to a sealed vessel at 1000 K. At equilibrium, there was 46.2% yield of CO₂ and total pressure was 3.4 atm.

- (i)** Write an expression for the equilibrium constant, K_p , for this reaction, including the units. [2]
- (ii)** Assuming that the gases behave ideally, explain why the initial total pressure was 3.4 atm. [1]
- (iii)** Calculate the equilibrium partial pressures of all species. [3]
- (iv)** Hence, calculate the value of K_p . [1]

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(d) Predict, with reasoning, the effect on yield and rate of reaction, if any, for WGSR, when

(i) pressure is increased [2]

(ii) temperature is increased [2]

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- (b) On titration, a solution containing 0.100 g of the acid **A**, RCO_2H , required a volume of 11.4 cm^3 of $0.100 \text{ mol dm}^{-3}$ NaOH to reach end point. Show that the molar mass of acid **A** is 87.7 g mol^{-1} and hence suggest the two possible structures of acid **A**. [3]

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- (c) A gaseous sample of 0.20 g of alkane **D** occupies a volume of $8.70 \times 10^{-5} \text{ m}^3$ at a temperature of 380 K and a pressure of $1.01 \times 10^5 \text{ Pa}$. Use these data to calculate the molar mass of alkane **D** and hence suggest its molecular formula. [2]

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- (d) Use your answer in (b) and (c) to give

(i) the structure of acid **B**, and [1]

(ii) one possible structure of alkane **E**. [1]

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[illegible]

9729/02/J1PROMO/20

Additional answer space

If you use the following pages to complete the answer to any question, the question number must be clearly shown.

[illegible]

